

Performance Testing Report Using Apache JMeter

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Testing Tool: Apache JMeter 5.6.3

Test Type: Performance Testing

Performance Testing of REST API Using Apache JMeter

Introduction

Performance Testing is a type of non-functional testing used to evaluate the speed, reliability, and stability of an application under different user loads.

In this task, Apache JMeter was used to simulate virtual users and analyze the performance of a REST API by measuring response time, throughput, and error percentage.

Testing Environment

Parameter	Details
Testing Tool	Apache JMeter 5.6.3
Java Version	JDK 21
Test Type	Performance Testing
Application/API Tested	JSONPlaceholder REST API
Endpoint Tested	GET /users
Number of Virtual Users	50
Operating System	Windows

Performance Test Plan

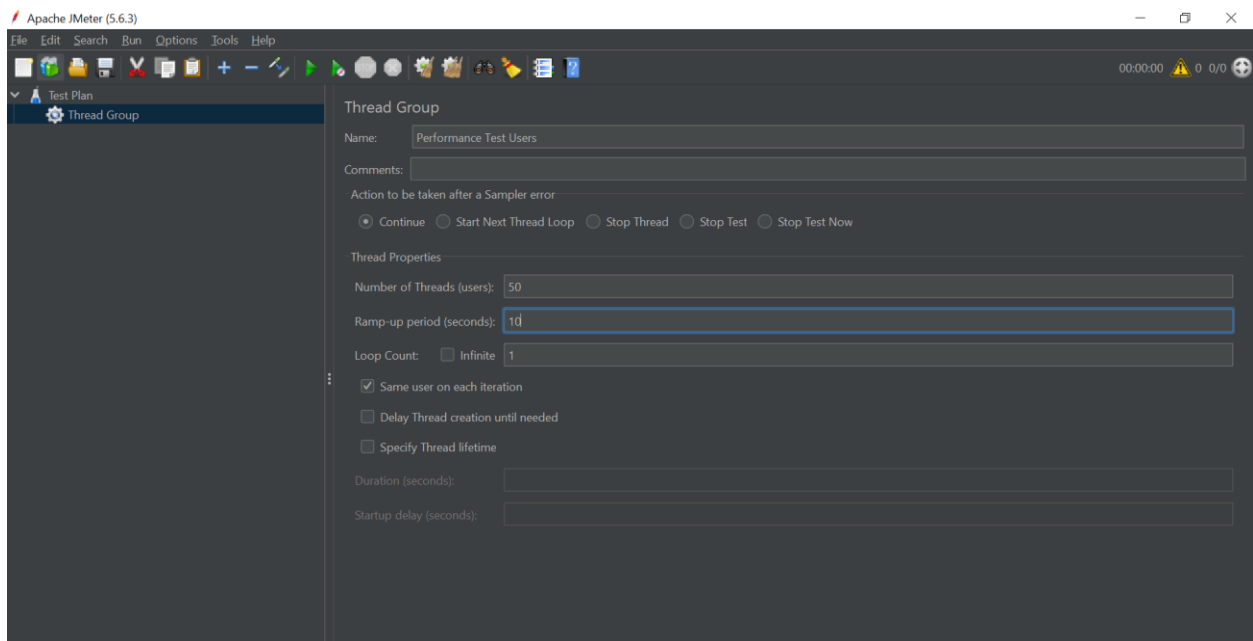
The objective of this performance test was to check how the API performs under multiple user requests.

The test was configured with 50 virtual users to analyze response time, throughput, and error percentage.

Load Configuration

Configuration	Value
Thread Group Name	Performance Test Users
Number of Threads (Users)	50
Ramp-up Period	10 seconds
Loop Count	1
Request Type	GET
Endpoint	/users

Thread Group Configuration

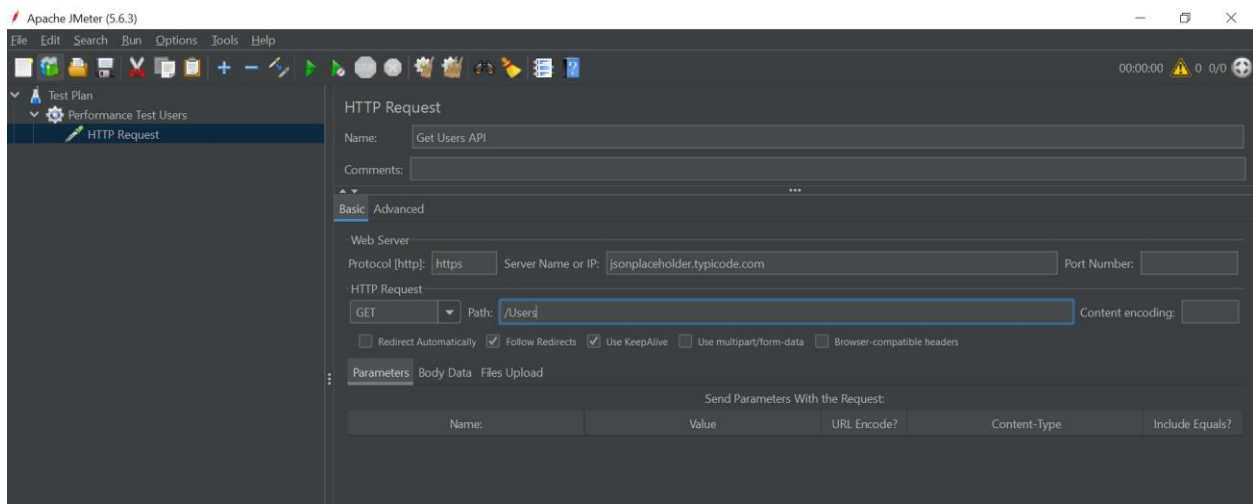


HTTP Request Configuration

An HTTP GET request was created in JMeter to test the users API endpoint.

Parameter	Value
Protocol	HTTPS
Server Name	jsonplaceholder.typicode.com
Path	/users
Method	GET

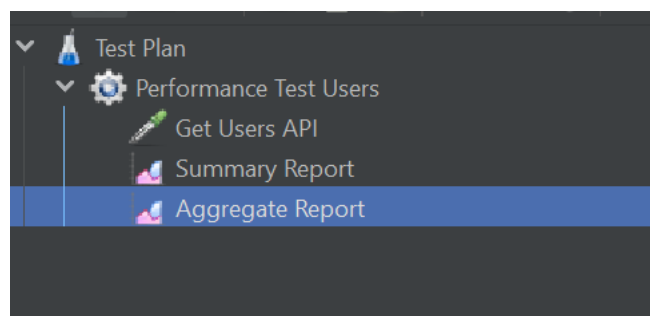
HTTP Request Configuration



Performance Test Execution

The performance test was executed using Apache JMeter with 50 virtual users. The test successfully generated performance reports including Summary Report and Aggregate Report.

JMeter Test Plan Structure



Performance Test Results

Test ID	Endpoint/Page	Users	Avg Response Time	Throughput	Error %	Status
PT-001	GET /users API	50	379 ms	4.9 requests/sec	0.00%	Pass

Aggregate Report Analysis

The Aggregate Report showed that the API handled all 50 requests successfully without any errors.

Metric	Result
Total Samples	50
Average Response Time	379 ms
Minimum Response Time	536 ms
Maximum Response Time	2123 ms
Error Percentage	0.00%
Throughput	4.9 requests/sec

Aggregate Report

Performance_Test.jmx (C:\Users\DELL\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Performance_Test.jmx) - Apache JMeter (5.6.3)

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Get Users ...	50	739	567	1260	1663	2123	536	2123	0.00%	4.9/sec	32.92	0.65
TOTAL	50	739	567	1260	1663	2123	536	2123	0.00%	4.9/sec	32.92	0.65

Summary Report Analysis

The Summary Report was generated to analyze the overall performance of the API.

The report showed successful execution of requests with zero errors and acceptable response time.

Summary Report

Performance_Test.jmx (C:\Users\DELL\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\Performance_Test.jmx) - Apache JMeter (5.6.3)

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Get Users API	50	739	536	2123	400.01	0.00%	4.9/sec	32.92	0.65	6863.7
TOTAL	50	739	536	2123	400.01	0.00%	4.9/sec	32.92	0.65	6863.7

☐ Include group name in label? ☒ Save Table Header

Error Analysis

During testing, all requests were successfully completed.

Parameter	Result
Total Requests	50
Failed Requests	0
Error Percentage	0.00%
Test Status	Passed

Performance Observations and Recommendations

1. The API successfully handled 50 virtual users without failure.
2. Error percentage remained 0%, showing stable performance.
3. Average response time was acceptable for the tested API.
4. Performance monitoring should be performed regularly before production deployment.
5. Server optimization and caching can further improve response time for higher user loads.

Conclusion:

The performance test was successfully completed using Apache JMeter. The tested API showed stable performance under 50 virtual users with zero errors and acceptable response time.